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CF-DWTA: Cascading-Failure Modeling and Multi-objective Optimization in Dense Target Scenarios

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Abstract:	Weapon target assignment aims to maximize combat effectiveness by optimizing assignment strategies under limited defense resource constraints. However, existing research often overlooks cascading failure effects in dense formation scenarios, resulting in significant deviations in calculating the damage probability against incoming targets. To address this issue, this article proposes a cascading-failure-based dynamic weapon target assignment (CF-DWTA) model, which ensures computational accuracy in dense target scenarios. To solve this model, an improved multi-objective particle swarm optimization algorithm, called GS-MOPSO, is proposed in this paper. This algorithm incorporates three performance improvement strategies: a group segmentation-based non-hierarchical fast non-dominated sorting strategy, a chaotic quasireflection elite mean learning initialization strategy, and an archive maintenance operator, aiming to balance convergence performance and computational efficiency. Simulation results illustrate that the proposed CF-DWTA model exhibits high computational accuracy. Furthermore, the GSMOPSO algorithm shows significant advantages in convergence, diversity, and runtime.

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Dear Editors,

This is Fuqiang Liu from Chongqing University. My co-authors and I have been developing application-oriented weapon target assignment (WTA) methods for real-time intelligent warfare. For dense target scenarios, this article derives the calculation formula for the cascading damage probability of targets and establishes a CF-DWTA model. To efficiently solve the model, we designed the GS-MOPSO algorithm, which effectively improves the efficiency and quality of solving the WTA problem by improving the population initialization, sorting strategies, and archive maintenance mechanisms. Since the published articles, (2024, 197: 110606) and (2023, 181: 109303) from *Computers & Industrial Engineering*, are highly related to solving the WTA problem and the research topic of our paper is also closely matched to it, we would like to submit it to *Computers & Industrial Engineering.*

I would like to declare on behalf of my co-authors that this work is our original research and has not been published previously; meanwhile, it will not be considered for publication elsewhere in whole or in part. We deeply appreciate your consideration of our manuscript. Please contact me if you have any questions.

Thank you very much for your generous help.

Yours faithfully,

Fuqiang Liu

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CF-DWTA: Cascading-Failure Modeling and Multi-objective Optimization in Dense Target Scenarios

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Highlights

CF-DWTA: Cascading-Failure Modeling and Multi-objective Optimization in Dense Target Scenarios

- The CF-DWTA model is proposed to alleviate the probability overestimation problem of Noisy-OR and LBP models in calculating cyclic probability graphs.
- An improved multi-objective particle swarm algorithm is designed to enhance the computational efficiency and convergence performance of the algorithm.
- The CF-DWTA model exhibits superior computational accuracy, and the GS-MOPSO demonstrates excellent computational efficiency and convergence.

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1. Introduction

As a core problem in military operations research, Weapon Target Assignment aims to maximize combat effectiveness by optimizing assignment strategies under limited defense resource constraints (Manne (1958)). With the rapid development of weapons and equipment, the WTA problem has become a research hotspot in the fields of military operations research and intelligent decision-making (Li et al. (2024); Bi et al. (2025)). To solve this problem, it is necessary to establish a reasonable WTA problem model and design efficient solving algorithms.

At present, research on the WTA model mainly focuses on enhancing its generalization and decision-making value by optimizing the diversified design of objectives and constraints. Maximizing destructive efficiency and minimizing resource consumption constitute the benchmark optimization objectives of the WTA model (Ma et al. (2021); Hocaoglu (2019)). In terms of constraints, the model has evolved from simple quantity limitations and discrete resource distributions to more complex logical constraints (Lamba et al. (2025)). For example, Tashakori et al. (2024) integrated communication topology and heterogeneity capabilities in drone routing. Si and Hao (2023) considered data loss constraints to ensure strike consistency in collaborative operations. The aforementioned research has promoted the development of WTA models, but it mainly considered static scenarios. To this end, researchers proposed the dynamic WTA model (DWTA) and defined it as a multi-stage static WTA, with the time dimension becoming its core factor. This change aims to improve the closeness of the DWTA model to dynamic combat scenarios.

With the development of artificial intelligence, swarm attacks and other dense formation tactics have become a dominant trend on modern battlefields. Although existing DWTA models provide solutions for the dynamic assignment problem to some extent (Arslan et al. (2025); Peng et al. (2024)), they ignore the effects of spatial damage propagation and detonation in dense formations, resulting in significant deviations in the calculated target damage probability. Therefore, researchers have begun to focus on cascading failure effects in dense formations (Yuan et al. (2025); Guo et al. (2019)), but existing methods struggle to achieve a balance between computational speed and accuracy. Specifically, Monte Carlo simulation has good computational accuracy, but its computational efficiency is relatively low for large-scale DWTA (Xin et al. (2011)). In contrast, probability graph models such as Noisy-OR and Loop Belief Propagation (LBP) have better computational efficiency, but cannot isolate redundant probabilities in the feedback loop, resulting in serious overestimation bias in the cyclic graph (Burkholz and Quackenbush (2021)). Therefore, developing a probability calculation model that balances efficiency and accuracy is a key challenge in current research.

In terms of solving algorithms, they are mainly divided into three categories: exact algorithms, deep reinforcement learning, and heuristic algorithms (Liu et al. (2025)). Accurate algorithms have good computational performance in small-scale scenarios, but as the scale of operations increases, they are prone to fall into the dilemma of exploding

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dimensions (Bertsimas and Paskov (2025); Lu and Chen (2021)). The emergence of deep reinforcement learning technology provides a new solving approach for the DWTA problem. For example, Luo et al. (2022) proposed the PODRL framework to achieve real-time solution of DWTA problem through improved deep Q-learning. To further enhance the quality of multi-objective decision-making, Peng et al. (2024) proposed a solution framework based on multi-agent strategy, which strengthens the convergence performance of the algorithm by introducing dynamic rewards. Although DRL exhibits excellent real-time performance in solving DWTA problem, it is limited by difficulties in data acquisition, poor convergence, and insufficient interpretability of decisions in practical applications (Li et al. (2024)). In contrast, heuristic algorithms are supposed to be an effective path for solving engineering optimization problem such as DWTA due to the strong convergence and generalization. Currently, researchers have achieved some accomplishments. For example, multi-objective particle swarm optimization algorithm (Yi et al. (2024)), genetic algorithm (Li et al. (2018)), improved artificial bee colony algorithm (Chang et al. (2018)), and improved firefly algorithm have been proposed successively. These studies have made outstanding contributions to enhancing the algorithm convergence and maintaining the diversity of solutions, providing a theoretical basis for the solution of DWTA. Especially, the multi-objective particle swarm algorithm has a simple structure and strong global search ability, which has significant advantages in solving DWTA problem (Chen et al. (2025); Zeng et al. (2025)).

However, the computational efficiency of heuristic algorithms is an urgent issue that needs to be addressed. Specifically, population size, the number of iterations, and non-dominated sorting are three major factors that determine the computational efficiency (Canpolat et al. (2024)). It is worth noting that population size and the number of iterations are controllable external factors. In contrast, non-dominated sorting is an internal mechanism of the algorithm; therefore, how to reduce redundant sorting operations at the logical level has become the key to improving its computational efficiency (Zhang et al. (2015)). Additionally, the convergence performance of an algorithm directly determines the quality of multi-objective decision-making; thus, a balance must be sought between its convergence performance and computational efficiency (Wu et al. (2023)). Based on these considerations, this article incorporates the time dimension, comprehensively considers the cascading damage effects between targets, constructs a DWTA model suitable for dense formation scenarios, and builds an efficient heuristic algorithm. The main contributions of this article are as follows:

1) A cascading-failure-based dynamic weapon target assignment model (CF-DWTA) is proposed. Based on the theory of cavity probability, this model derives the formula for calculating the damage probability under target cascade failure, and alleviates the problem of overestimation of probabilities in the calculation of cyclic probability graphs for Noisy-OR and LBP models.

2) An improved particle swarm algorithm named GS-MOPSO is designed. The sorting strategy, population initialization strategy, and external archiving mechanism of traditional MOPSO are strengthened to improve the computational efficiency and convergence performance of the algorithm, ensuring its application in dynamic environments.

3) The effectiveness of the proposed CF-DWTA model in terms of computational accuracy is verified through 12 sets of test cases. Meanwhile, simulation experiments in six different scale scenarios further demonstrate that GS-MOPSO has significant advantages in convergence, diversity, and runtime when solving DWTA problem.

The remainder of this article is organized as follows. Section 2 presents the problem definition of the CF-DWTA model. In Section 3, the GS-MOPSO algorithm is proposed to solve the CF-DWTA problem. Subsequently, Section 4 details the experimental analysis of the proposed approach , and Section 5 concludes the article.

2. Problem Definition

Although the existing DWTA model incorporates the time dimension into the decision-making process, it ignores the potential spread of damage caused by the destruction of incoming targets, resulting in calculation deviations of damage probability in dense formation scenarios. To enhance the adaptability of DWTA to actual combat environments, this paper proposes the CF-DWTA model based on the benchmark DWTA model and designs an EPL-AT strategy for efficiently solving the cascading damage probability. The principle of the proposed CF-DWTA model is shown in Fig. 1 and the symbol descriptions of the CF-DWTA model are shown in Table 1. Fig. 1 illustrates the computational mechanism of the CF-DWTA model. Specifically, the comprehensive damage probability of the target is calculated from the direct damage probability (DDP) generated by weapon strikes and the indirect damage probability (IDP) caused by neighboring target destruction. This paper proposes the EPL-AT strategy to efficiently calculate the indirect probability under the influence of cascading damage. This strategy adaptively truncates the cascading propagation

Table 1
Parameters Representation and Description of CF-DWTA Model

Parameters	Description	Dimension
$\mathcal{H}, n$	The set of incoming targets and its total number	Set / Scalar
$\mathcal{W}, m$	The set of defensive weapons and its total number	Set / Scalar
T	The total number of interception stages in the dynamic scenario	Scalar
$\mathbf{P}_{\text{wea}}$	The direct damage probability matrix of defensive weapons	$m \times n$
$\mathbf{M}_{\text{cas}}$	The single-stage cascade failure probability adjacency matrix	$n \times n$
$\boldsymbol{p}^{\text{DDP}}$	The direct damage probability vector of targets (direct attack)	$n \times 1$
$\boldsymbol{p}^{\text{IDP}}$	The indirect damage probability vector of targets (cascade effect)	$n \times 1$
$\boldsymbol{p}^{\mathcal{H}}$	The final comprehensive damage probability vector of targets in incoming targets $\mathcal{H}$	$n \times 1$
$p_k^{\mathcal{H}-\{j\}}$	The cavity probability (probability of incoming target k being damaged when incoming target j is removed)	Scalar
$\boldsymbol{c}$	The operational cost vector of defensive weapons	$m \times 1$
$\boldsymbol{e}$	The available ammunition quantity vector of defensive weapons	$m \times 1$
$\boldsymbol{\lambda}$	The average flight speed vector of defensive weapons	$m \times 1$
$\boldsymbol{r}$	The preparation time vector of defensive weapons	$m \times 1$
$\boldsymbol{v}$	The tactical threat value vector of incoming targets	$n \times 1$
$O_{\max}$	The maximum truncation order (EMCFPL threshold)	Scalar
τ	The probability truncation threshold for the EPL-AT strategy	Scalar
$P_j^{\min}, P_j^{\max}$	The lower and upper bounds of expected damage probability for target j	Scalar
$\mathbf{X}$	The Weapon-target assignment decision matrix	$m \times n$

paths, eliminates redundant probability coupling terms, and iteratively updates the comprehensive damage probability of the target.

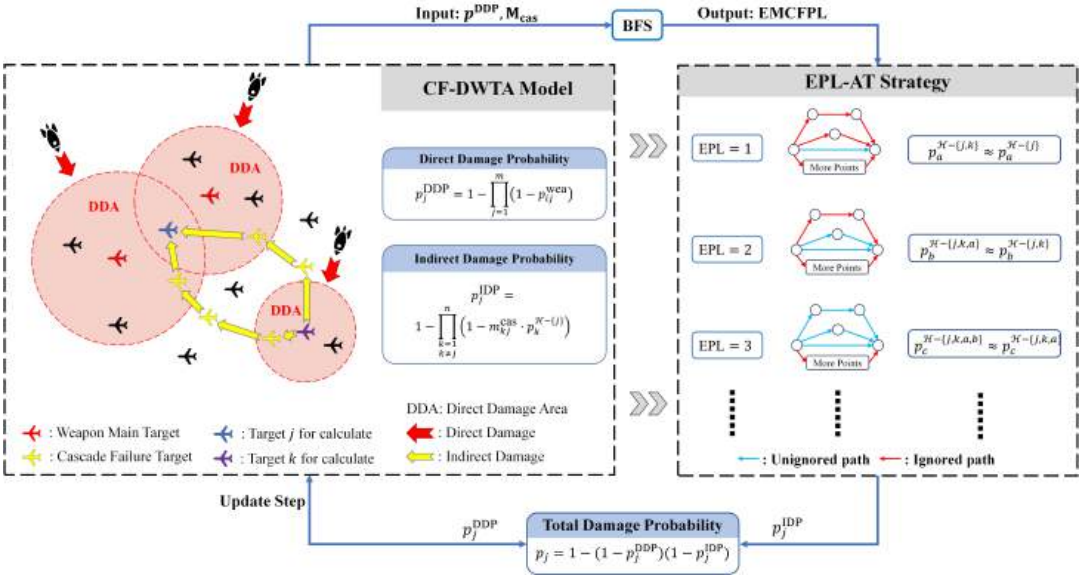


Figure 1: CF-DWTA model and EPL-AT strategy diagram

2.1. Basic Assumptions

In dense formation scenarios, the destruction of incoming targets may affect neighboring areas and trigger cascading effects. Therefore, the target damage probability calculation in the CF-DWTA model considers two aspects. On the one hand, it accounts for the direct damage inflicted by defensive weapons; on the other hand, it incorporates the indirect impact caused by the destruction of neighboring targets. Accordingly, to simplify the model, this article proposes the following definitions and assumptions:

1) The direct damage probability matrix $\mathbf{P}_{\text{wea}} = [p_{ij}^{\text{wea}}]_{m \times n}$ is defined to represent the direct damage probability of the defensive weapon to the target, where p_{ij}^{wea} represents the damage probability of the i-th defensive weapon intercepting the j-th incoming target.

2) A cascading failure matrix $\mathbf{M}_{\text{cas}} = [m_{kj}^{\text{cas}}]_{n\times n}$ is defined to represent the indirect damage effect between neighboring targets, where m_{kj}^{cas} represents the conditional probability of target j suffering indirect damage under the condition that target k is damaged. It should be pointed out that indirect destructive effects only exist between specific neighboring targets, therefore matrix $\mathbf{M}_{\text{cas}}$ has sparsity. Moreover, this article assumes that $\mathbf{M}_{\text{cas}}$ has characterized the spatial coupling effects caused by multidimensional factors such as target type and spatial location at the macro probability level.

3) It is assumed that the incoming targets are simultaneously hit by defensive weapons, i.e. ignoring the time difference in damage between each incoming target during a single attack.

2.2. Calculation of Target Destruction Probability

The damage probability of incoming targets consists of two parts, the DDP caused by direct strikes from defensive weapons and the IDP caused by the destruction of neighboring targets. Therefore, the calculation formula for the damage probability of the incoming target j is given as follows:

$$p_j = 1 - (1 - p_j^{\text{DDP}}) \cdot (1 - p_j^{\text{IDP}}) \tag{1}$$

where p_j^{DDP} represents the direct damage probability of target j, and p_j^{IDP} represents the indirect damage probability of target j.

Since the attack events of each defensive weapon are independent, the DDP can be calculated using the standard Noisy-OR model, as follows:

$$p_j^{\text{DDP}} = 1 - \prod_{i=1}^{m}(1 - p_{ij}^{\text{wea}}) \tag{2}$$

where m is the number of defensive weapons, and p_{ij}^{wea} is the direct damage probability of defensive weapon i against incoming target j.

In traditional damage probability calculation models, the indirect damage probability p_j^{IDP} is often simplified to the standard Noisy-OR model, as follows:

$$p_j^{\text{IDP}} = 1 - \prod_{\substack{k=1\\k\neq j}}^{n}(1 - m_{kj}^{\text{cas}} \cdot p_k) \tag{3}$$

where m_{kj}^{cas} is the indirect damage probability of target k against target j, and p_k is the total damage probability of target k.

Due to the fact that Eq. (3) ignores the indirect damage contribution caused by the target j in the damage probability p_j^{IDP}, the damage probability of target j is overestimated. In response to this issue, this article introduces the concept of cavity probability from statistical physics, and modifies p_k in Eq. (3) to cavity probability $p_k^{\mathcal{H}-\{j\}}$. $p_k^{\mathcal{H}-\{j\}}$ represents the probability of damage to target k after removing target j from the current set of incoming targets $\mathcal{H}$. By substituting the revised Eq. (3) into Eq. (1), one can obtain,

$$p_j^{\mathcal{H}} = 1 - (1 - p_j^{\text{DDP}}) \cdot \prod_{\substack{k=1\\k\neq j}}^{n}(1 - m_{kj}^{\text{cas}} \cdot p_k^{\mathcal{H}-\{j\}}) \tag{4}$$

where $p_j^{\mathcal{H}}$ represents the total damage probability of incoming target j, and $p_k^{\mathcal{H}-\{j\}}$ represents the cavity probability of target k.

To precisely solve for $p_k^{\mathcal{H}-\{j\}}$, it is necessary to further trace and eliminate deeper causal propagation paths, thereby forming the following system of recursive equations:

$$\begin{cases} p_j^{\mathcal{H}} = 1-(1-p_j^{\mathrm{DDP}})\prod\limits_{\substack{k=1\\k\neq j}}^{n}(1-m_{kj}^{\mathrm{cas}}\cdot p_k^{\mathcal{H}-\{j\}}) \\ p_k^{\mathcal{H}-\{j\}} = 1-(1-p_k^{\mathrm{DDP}})\prod\limits_{\substack{a=1\\a\neq j,k}}^{n}(1-m_{ak}^{\mathrm{cas}}\cdot p_a^{\mathcal{H}-\{j,k\}}) \\ p_a^{\mathcal{H}-\{j,k\}} = 1-(1-p_a^{\mathrm{DDP}})\prod\limits_{\substack{b=1\\b\neq j,k,a}}^{n}(1-m_{ba}^{\mathrm{cas}}\cdot p_b^{\mathcal{H}-\{j,k,a\}}) \\ \quad\vdots \\ p_r^{\{r,n\}} = 1-(1-p_r^{\mathrm{DDP}})(1-m_{nr}^{\mathrm{cas}}\cdot p_n^{\{n\}}) \\ p_n^{\{n\}} = p_n^{\mathrm{DDP}} \end{cases} \tag{5}$$

However, since the computational complexity of Eq. (5) is $O(n!)$, the computational cost will increase exponentially when the number of incoming targets is large. Therefore, it is necessary to design an efficient method to improve computational efficiency. To address this, we design a fast solution strategy, hereafter referred to as EPL-AT.

2.3. Effective Path-Length Adaptive Truncation Strategy

To clarify the solution structure of EPL-AT, it is necessary to explain the principle of cascading failure. Specifically, the impact of cascading failures decays with increasing the propagation path length, and cascading effects with longer propagation paths can usually be ignored. Besides, due to the fact that a single target only indirectly damages its neighboring targets, the cascading failure adjacency matrix $\mathbf{M}_{\mathrm{cas}}$ exhibits sparse characteristics. Based on this, one can approximately solve Eq. (5) by limiting the maximum propagation path length, thereby reducing computational complexity.

Taking second-order truncation as an example (considering at most two secondary coupling effects), we define the propagation path as $j \to k \to a$. Therefore, Eq. (5) can be approximated as

$$\begin{cases} p_j^{\mathcal{H}} = 1-(1-p_j^{\mathrm{DDP}})\prod\limits_{\substack{k=1\\k\neq j}}^{n}(1-m_{kj}^{\mathrm{cas}}\cdot p_k^{\mathcal{H}-\{j\}}) \\ p_k^{\mathcal{H}-\{j\}} = 1-(1-p_k^{\mathrm{DDP}})\prod\limits_{\substack{a=1\\a\neq j,k}}^{n}(1-m_{ak}^{\mathrm{cas}}\cdot p_a^{\mathcal{H}-\{j,k\}}) \\ p_a^{\mathcal{H}-\{j,k\}} \approx 1-(1-p_a^{\mathrm{DDP}})\prod\limits_{\substack{b=1\\b\neq j,k,a}}^{n}(1-m_{ba}^{\mathrm{cas}}\cdot p_b^{\mathcal{H}-\{j,k\}}) \end{cases} \tag{6}$$

This method approximates the deep cavity probability $p_b^{\mathcal{H}-\{j,k,a\}}$ in Eq. (5) as $p_b^{\mathcal{H}-\{j,k\}}$, thereby ignoring cascading propagation effects beyond three nodes. To capture cascading effects beyond three nodes, higher-order truncation strategies are required. However, cascading failure effects vary among different targets, and high-order truncation can lead to computational redundancy. To address this issue, EPL-AT uses the cascading failure adjacency matrix $\mathbf{M}_{\mathrm{cas}}$ to calculate the effective maximum cascading-failure path length (EMCFPL) between different targets, thereby adaptively selecting the truncation order for each target.

Specifically, we first select two targets and set the EMCFPL threshold and probability truncation threshold, and then use the breadth-first search (BFS) algorithm to search for cascading failure paths between these two targets. When

the path propagation probability is below the threshold, the path is too long, or the target is reached, the search process is stopped, and the truncation order of the target is determined based on the longest path length. This process is repeated until the truncation order of each pair of targets is determined, and the cascading damage probability of the targets is calculated. The implementation details of EPL-AT are shown in Algorithm 1. The specific process is as follows: Firstly, BFS is utilized to obtain all effective cascading failure paths between incoming targets, and EMCFPL is selected as the truncation order O. Then, Eq. (5) is simplified based on the truncation order O, and the cavity probability is iteratively calculated for each pair of incoming targets. Finally, the cavity probability is combined with the DDP to obtain the comprehensive damage probability $\boldsymbol{p}^{\mathcal{H}}$ of the incoming target.

Algorithm 1 EPL-AT

Input: Direct damage probability $\boldsymbol{p}^{\mathrm{DDP}}$, Adjacency matrix $\mathbf{M}_{\mathrm{cas}}$, Probability threshold τ, Max truncation order $O_{\max}$
Output: Final comprehensive damage probability $\boldsymbol{p}^{\mathcal{H}}$

Initialize truncation order matrix $\boldsymbol{\mathcal{O}} \leftarrow \mathbf{1}$
for each target $j \in \{1, \ldots, N\}$ **do**
  Extract valid cascading paths $\mathcal{P}_j \leftarrow \mathrm{BFS}(\mathbf{M}_{\mathrm{cas}}, j, O_{\max}, \tau)$
  for each path $p \in \mathcal{P}_j$ ending at target k **do**
    $\boldsymbol{\mathcal{O}}[j,k] \leftarrow \max(\boldsymbol{\mathcal{O}}[j,k], \mathrm{length}(p))$
  end for
end for
Initialize cavity probability matrix $\boldsymbol{C} \leftarrow \boldsymbol{p}^{\mathrm{DDP}}$
repeat
  $\boldsymbol{C}_{\mathrm{prev}} \leftarrow \boldsymbol{C}$
  for each target k and isolated target j $(j \neq k)$ **do**
    Calculate truncated message $msg_{a\to k}$ from valid neighbors a using $\boldsymbol{C}_{\mathrm{prev}}$ and order $\boldsymbol{\mathcal{O}}[j,k]$
    $\boldsymbol{C}[k,j] \leftarrow 1 - (1 - \boldsymbol{p}^{\mathrm{DDP}}[k]) \prod_{a\neq j} \left(1 - \mathbf{M}_{\mathrm{cas}}[a,k] \cdot msg_{a\to k}\right)$
  end for
until $\max |\boldsymbol{C} - \boldsymbol{C}_{\mathrm{prev}}| < \epsilon$
for $j = 1$ **to** n **do**
  $\boldsymbol{p}^{\mathcal{H}}[j] \leftarrow 1 - (1 - \boldsymbol{p}^{\mathrm{DDP}}[j]) \prod_k \left(1 - \mathbf{M}_{\mathrm{cas}}[k,j] \cdot \boldsymbol{C}[k,j]\right)$
end for
return $\boldsymbol{p}^{\mathcal{H}}$

2.4. CF-DWTA Model

CF-DWTA divides the entire interception process into multiple consecutive combat stages, and the weapon target assignment scheme for each stage can be dynamically adjusted based on the results of the previous stage to adapt to the constantly changing battlefield environment and incoming target status. Specifically, after the attack in stage t is completed, the system observes the remaining set of incoming targets and uses it as the set of incoming targets $\mathcal{H}$ for stage $t+1$.

In the DWTA scenario, the defense effect is characterized by the sum of the remaining threats to the incoming targets at each stage, which is expressed as follows:

$$f_1(\mathbf{X}) = \sum_{t=1}^{T} \sum_{j=1}^{n} v_j(t)(1 - p_j^{\mathcal{H}}(t, \mathbf{X})) \tag{7}$$

where $p_j^{\mathcal{H}}(t, \mathbf{X})$ represents the comprehensive damage probability of incoming target j at stage t; v_j is the threat level of incoming target j; T is the total number of stages for DWTA.

In order to enhance the sustained combat capability of the interception system, it is necessary to minimize the consumption of defense resource costs. The specific expression is as follows:

$$f_2(\mathbf{X}) = \sum_{t=1}^{T} \sum_{j=1}^{n} \sum_{i=1}^{m} c_i \cdot x_{ij}(t) \tag{8}$$

where c_i represents the cost of executing one interception task by the i-th defensive weapon; $x_{ij}(t)$ represents the quantity of ammunition allocated by weapon system i to target j at stage t.

To ensure a basic defensive baseline, each incoming target must be assigned at least one defensive firepower unit for interception:

$$\sum_{i=1}^{m} x_{ij}(t) \geq 1, \quad \forall j \in \{1, 2, \ldots, n\} \tag{9}$$

The total quantity of ammunition expended by each defensive weapon system at the current stage should not exceed its available ammunition limit:

$$\sum_{j=1}^{n} x_{ij}(t) \leq e_i(t), \quad \forall i \in \{1, 2, \ldots, m\} \tag{10}$$

where $e_i(t)$ is the available ammunition quantity of the i-th defensive weapon at stage t.

To avoid excessive damage to high-value targets and insufficient damage to low value targets, the degree of damage to target j should be controlled within a reasonable range, as follows:

$$P_j^{\min} \leq 1 - \prod_{t=1}^{T} (1 - p_j^{\mathcal{H}}(t, \mathbf{X})) \leq P_j^{\max}, \quad \forall j \in \{1, 2, \ldots, n\} \tag{11}$$

where $P_j^{\min}$ and $P_j^{\max}$ are the lower and upper bounds of the damage probability for incoming target j, respectively.

Due to limited fire channels, the total ammunition allocated by weapon system i at stage t cannot exceed its simultaneous engagement capacity C_i:

$$\sum_{j=1}^{n} x_{ij}(t) \leq C_i, \quad \forall i \in \{1, \ldots, m\} \tag{12}$$

Since the CF-DWTA is a multi-stage dynamic process, the available defensive resources undergo continuous consumption. The state transition equation for the remaining ammunition of weapon system i is formulated as follows:

$$e_i(t+1) = e_i(t) - \sum_{j=1}^{n} x_{ij}(t), \quad \forall i \in \{1, \ldots, m\}, \forall t \in \{1, \ldots, T-1\} \tag{13}$$

where $e_i(1)$ is initialized to the maximum carrying capacity of the i-th weapon system.

In summary, the CF-DWTA model proposed in this article is represented as follows

$$\begin{cases} \min\limits_{\mathbf{X}} \boldsymbol{F}(\mathbf{X}) = [f_1(\mathbf{X}), f_2(\mathbf{X})]^{\mathrm{T}} \\ \text{s.t.} \quad \text{Eqs.}(9) - (13) \end{cases} \tag{14}$$

3. Methodology

MOPSO, as a classic algorithm for solving engineering optimization problems, has strong global search capabilities (Li (2023); He et al. (2022)), which provide significant advantages for solving the DWTA problem. However, traditional MOPSO usually adopts a multi-layer Pareto solution set partitioning and a traversal comparison strategy based on dominance relationships across the entire population, which significantly increases its computational cost in large-scale decision spaces. Moreover, existing archive maintenance mechanisms ignore the potential value of some crowded particles, thereby weakening the algorithm's ability to maintain population diversity.

In response to the above issues, this article proposes GS-MOPSO, whose computational framework is illustrated in Fig. 2. Specifically, an improved fast sorting strategy, GS-NNS, is proposed to enhance the computational efficiency of the algorithm. At the same time, two improvement strategies are designed to enhance the convergence performance of the algorithm, namely Chaotic quasi reactive elite mean learning (CQREML) and crowding distance sigmoid variant mapping for archive maintenance (CD-SVAM). The three strategies work together to maintain a balance between the algorithm's computational efficiency and convergence performance.

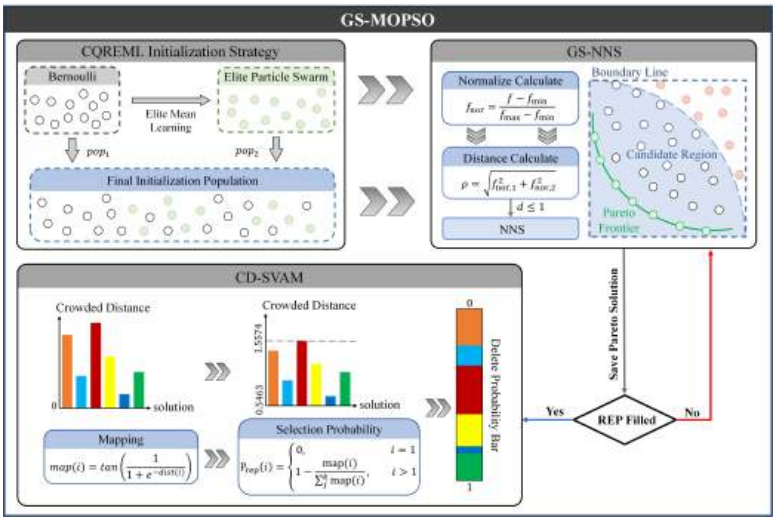


Figure 2: Flowchart and algorithmic framework of the proposed GS-MOPSO.

3.1. CQREML

The quality of population initialization has a direct impact on the search and convergence capabilities of the algorithm, and the original random initialization strategy of MOPSO struggles to ensure population diversity. In response to this issue, this article proposes the CQREML strategy, which combines Bernoulli chaotic mapping and a quasi-reflective elite mean learning mechanism to enhance population diversity.

Firstly, the initial population is generated using Bernoulli chaotic mapping, as shown in Eq. (15). This method can evenly distribute the population within the feasible region.

$$\beta_{t+1} = \begin{cases} \frac{\beta_t}{1-\alpha}, & 0 < \beta_t \leq 1-\alpha \\ \frac{\beta_t-(1-\alpha)}{\alpha}, & 1-\alpha < \beta_t < 1 \end{cases} \tag{15}$$

where β_t denotes the chaotic variable produced at the t-th iteration, and $\alpha \in (0, 1)$ is the parameter governing the mapping dynamics. The particle swarm generated by the Bernoulli mapping within the decision space is designated as the exploratory population pop_1.

Subsequently, to compensate for the blindness of chaotic mapping, this paper introduces a quasi-reflective elite mean learning mechanism. This mechanism generates more guided inverse solutions by calculating the average position of elite particles. The average position of elite particles in pop_1 is calculated as follows,

$$\boldsymbol{x}^{\mathrm{po}} = \frac{\sum_{v=1}^{V} \boldsymbol{x}_v^p}{V} \tag{16}$$

where $\boldsymbol{x}_v^p$ denotes the position vector of the v-th best particle in pop_1, and V represents the size of the elite set. Subsequently, opposition-based solutions are generated based on this mean position:

$$\boldsymbol{x}_v^* = \begin{cases} \boldsymbol{x}^{\mathrm{po}} + r(\boldsymbol{x}_v - \boldsymbol{x}^{\mathrm{po}}), & \boldsymbol{x}_v > \boldsymbol{x}^{\mathrm{po}} \\ \boldsymbol{x}_v + r(\boldsymbol{x}^{\mathrm{po}} - \boldsymbol{x}_v), & \boldsymbol{x}_v < \boldsymbol{x}^{\mathrm{po}} \end{cases} \tag{17}$$

where $\boldsymbol{x}_v$ denotes the original position vector of a particle in pop_1, $\boldsymbol{x}_v^*$ represents the newly generated position vector, and $r \in (0, 1)$ is a uniformly distributed random number. The set consisting of all $\boldsymbol{x}_v^*$ is defined as the guiding population pop_2. Finally, the population pop_1 is merged with population pop_2 to form the final initial population.

3.2. GS-NNS

The fast non-dominated sorting (FNDS) method is the core sorting strategy for MOPSO in handling multi-objective problems. However, FNDS requires sorting all solutions and forces all individuals to undergo pairwise comparisons,

which results in sorting redundancy and decreased computational efficiency. This article proposes the GS-NNS strategy to address this issue. This strategy consists of two parts: non-hierarchical fast non-dominated sorting (NNS) and group segmentation (GS), aiming to improve the algorithm's computational efficiency by simplifying the sorting logic.

Specifically, the NNS method outputs only the solution of the first frontier based on FNDS, while terminating the sorting process of all subsequent levels to avoid repeated comparisons and unnecessary advantage comparisons of particles. Furthermore, GS-NNS utilizes the cropping mechanism of GS to pre-screen particles with poor fitness values, thereby optimizing sorting speed. Firstly, the fitness values of the two objective functions are separately normalized as follows

$$f_{\text{nor}} = \frac{f - f_{\min}}{f_{\max} - f_{\min}} \tag{18}$$

where f_{nor} denotes the normalized fitness value, whose range is $[0, 1]$; f represents the original fitness derived from the CF-DWTA model; and $f_{\max}$ and $f_{\min}$ represent the maximum and minimum fitness values within the current solution set, respectively.

Secondly, the Euclidean distance d_i from f_{nor} to the origin is defined as the filtering distance, where d_i is represented as follows

$$d_i = \sqrt{(f^i_{\text{nor},1})^2 + (f^i_{\text{nor},2})^2} \tag{19}$$

Finally, the region $d \leq 1$ is defined as a candidate region, and the NNS operation is only performed on particles located in this region. The implementation details of GS-NNS are shown in Algorithm 2.

Algorithm 2 GS-NNS

Input: Population P, Number of individuals NP, Number of objective function values M
Output: External archive REP

for $i = 0$ **to** $NP - 1$ **do**
  for $k = 0$ **to** $M - 1$ **do**
    $f^{\text{norm}}_{i,k} \leftarrow \text{Normalize}(f_{i,k})$
  end for
end for
$D_{\text{norm}} \leftarrow$ initial distance
$P_{\text{GS}} \leftarrow P$
for $i = 0$ **to** $NP - 1$ **do**
  if f_i is out of target range **then**
    $P_{\text{GS}} \leftarrow P_{\text{GS}} \setminus \{i\}$ // Delete from GS population
  end if
end for
for $i = 0$ **to** $|P_{\text{GS}}| - 1$ **do**
  for $k = 0$ **to** $|P_{\text{GS}}| - 1$ **do**
    if $i \neq k$ **and** $P_{\text{GS}}[i]$ is dominated by $P_{\text{GS}}[k]$ **then**
      break
    end if
  end for
  if $P_{\text{GS}}[i]$ is not dominated **then**
    $REP \leftarrow REP \cup \{P_{\text{GS}}[i]\}$ // Add to Pareto set
  end if
end for
return REP

3.3. CD-SVAM

The external archive (EA) is used to store Pareto frontier solution sets and provide globally optimal particles, but its size should be limited to avoid affecting the algorithm's performance. Furthermore, traditional maintenance strategies

based on crowding distance only retain particles with large crowding distances, which limits population diversity and evolutionary potential. To address this issue, this article proposes an archive maintenance operator called CD-SVAM. This operator consists of two parts: EA crowding distance calculation and probabilistic maintenance based on a Sigmoid variant mapping. This operator first calculates the crowding distances of all particles in the EA, and converts them into corresponding retention probabilities through the Sigmoid variant mapping, so that particles with larger crowding distances have a higher chance of being selected. When the EA size exceeds the limit, the operator prioritizes retaining particles with good convergence and diversity, while retaining some particles with smaller crowding distances with a smaller probability to maintain the diversity and evolutionary potential of the population.

Firstly, when the number of particles in the EA exceeds the upper limit, the crowding distances of all particles in the archive are calculated as follows

$$dist(i) = \sum_{m=1}^{M} \frac{f_m(i+1) - f_m(i-1)}{f_m^{\max} - f_m^{\min}} \tag{20}$$

where M denotes the total number of objective functions; m represents the objective function index; and $f_m^{\max}$ and $f_m^{\min}$ represent the maximum and minimum values of the m-th objective function, respectively.

Secondly, the Sigmoid variant function is used to map the crowding distance of all particles in EA, as shown below,

$$map(i) = \tan\left(\frac{1}{1 + e^{-dist(i)}}\right) \tag{21}$$

Then, the weight proportion P_{rep} of each particle after mapping can be calculated and defined as the particle's selection probability.

$$P_{\text{rep}}(i) = \begin{cases} 0, & i = 1 \\ 1 - \frac{map(i)}{\sum_{j=1}^{k} map(j)}, & i > 1 \end{cases} \tag{22}$$

Subsequently, one can select the particle i to be updated and generate the corresponding probability interval ℓ_i based on the selection probability of particle i.

$$\ell_i = \left[\sum_{j=1}^{i-1} P_{\text{rep}}(j), \sum_{j=1}^{i} P_{\text{rep}}(j)\right] \tag{23}$$

Finally, a random number *rand* that follows a uniform distribution of [0,1] is generated. If rand falls within the interval ℓ_i, particle i is deleted. Repeat the above selection process until the number of particles in the EA is reduced to the preset capacity limit.

4. Experiments and Analysis

In this section, numerical experiments are conducted to verify the effectiveness of the proposed algorithm. Due to the sensitive nature of the WTA problem, there is currently a lack of publicly available standard datasets. Therefore, based on the simulation framework proposed in existing research (Chang et al. (2023); Ma et al. (2021); Tashakori et al. (2024)), this paper designs six defensive combat scenarios of different scales. A test case generator is used to generate key parameters such as target threat levels and target damage thresholds. For algorithm comparison, this paper selects classic algorithms closely related to the WTA problem as well as several multi-objective optimization algorithms. The seven point average distance, hypervolume, and runtime are applied to evaluate the convergence, diversity, and computational efficiency of the algorithms, respectively. Furthermore, the Wilcoxon rank-sum test and the Friedman test are used to statistically analyze the simulation results to verify the robustness of the proposed algorithm. All numerical experiments are conducted on a workstation equipped with a 13th Gen Intel Core i9-13900HX processor and 16 GB of RAM. All algorithms are implemented and evaluated using MATLAB R2025b.

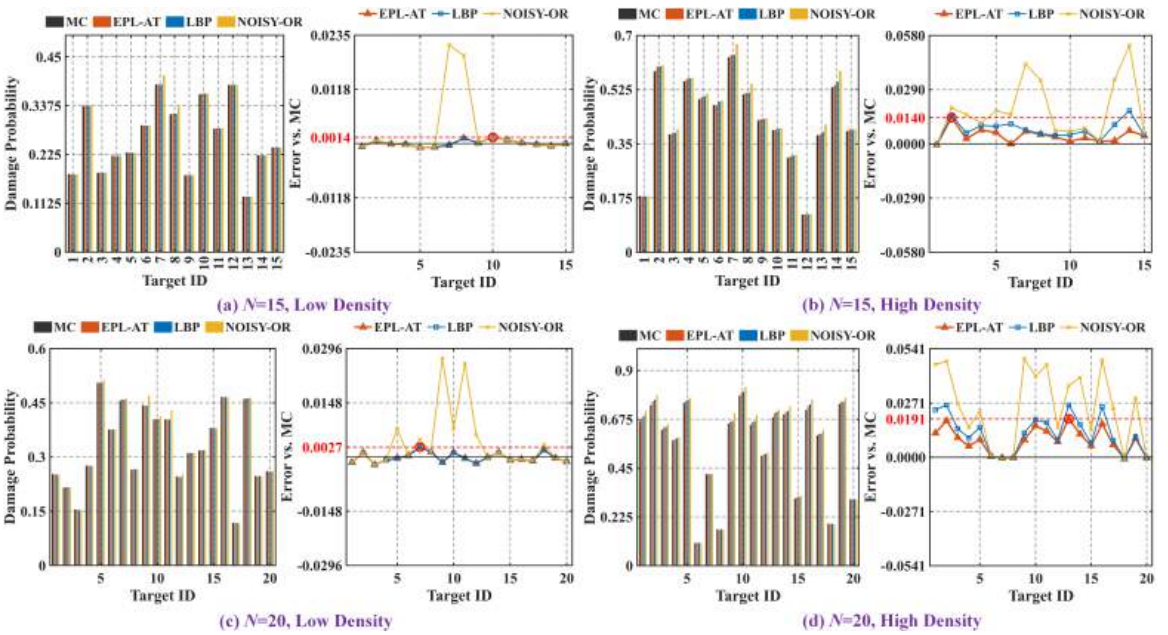


Figure 3: A comprehensive evaluation of damage probability estimations and calculation errors under varying target quantities (N) and network densities. Each subfigure presents the estimated damage probabilities of individual targets (left) and the absolute errors relative to the Monte Carlo ground truth (right). The proposed EPL-AT is compared with LBP and the standard Noisy-OR model, using 50,000 Monte Carlo simulations as the baseline.

4.1. Verification of EPL-AT Calculation Accuracy and Efficiency

To evaluate the computational accuracy and efficiency of EPL-AT under different numbers of targets and network densities, this paper combines the number of targets and network densities (high and low densities) to establish 12 sets of test cases. For each case, the accuracy and calculation error of various algorithms in estimating target damage probability are evaluated by comparing EPL-AT with LBP and Noisy-OR, using the results of 50000 MC simulations as a benchmark. The estimated damage probability of a single target and its absolute error relative to the MC benchmark (hereinafter referred to as estimation error) are used as evaluation metrics.

Figs. 3-5 show the comparison results of all methods. The results indicate that in most cases, the estimation error of EPL-AT is smaller than that of the LBP algorithm and the Noisy OR model. Particularly in high-density network scenarios, the advantages of EPL-AT are more pronounced. Fig. 6 shows a radar chart of the computational efficiency for all methods. The results indicate that although the computation time of EPL-AT is longer than that of LBP and Noisy-OR, the overall difference is not significant. Compared to MC, EPL-AT has a significant advantage in computational efficiency.

4.2. GS-NNS Effectiveness Verification

As GS-NNS is the core strategy for improving computational efficiency, it is necessary to examine its contribution to GS-MOPSO. To this end, this paper uses MOPSO as the basic framework and replaces its original sorting strategy with MNDS, CS (Wang and Yao (2013)), NNS, and GS-NNS, thus forming four comparative variant algorithms. These variants are then compared with the complete GS-MOPSO. All algorithms are run independently 30 times. It should be noted that the GS-NNS variant fully inherits all other computing components of MOPSO, while GS-MOPSO is the complete algorithm that integrates GS-NNS and all other improved strategies proposed in this paper. This design ensures that the improvement in algorithm performance is entirely derived from the optimization of the sorting strategy, rather than interference from other factors.

Fig. 7 shows the runtime comparison results of all algorithms across six scenarios of different scales. The results indicate that the proposed GS-NNS strategy significantly outperforms traditional sorting methods in terms of computational efficiency. In addition, GS-MOPSO, which integrates all proposed strategies, not only maintains a fast solving speed but also exhibits significant operational robustness.

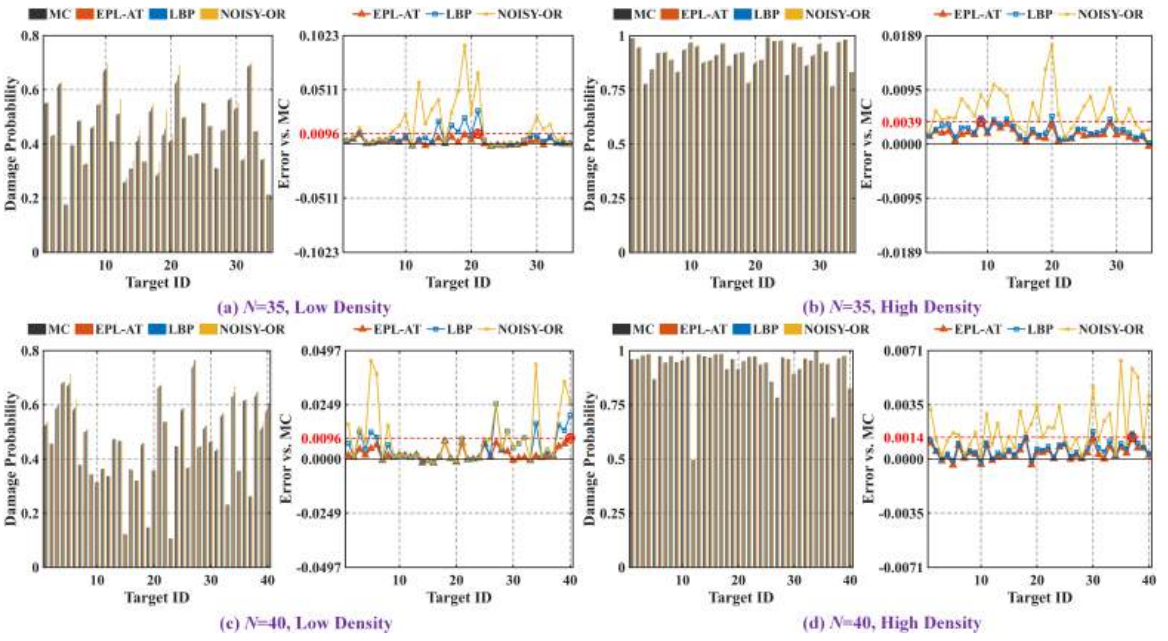


Figure 4: Comparison of damage probability estimations and absolute errors for medium-scale scenarios ($N = 35$ and 40) under low and high network densities.

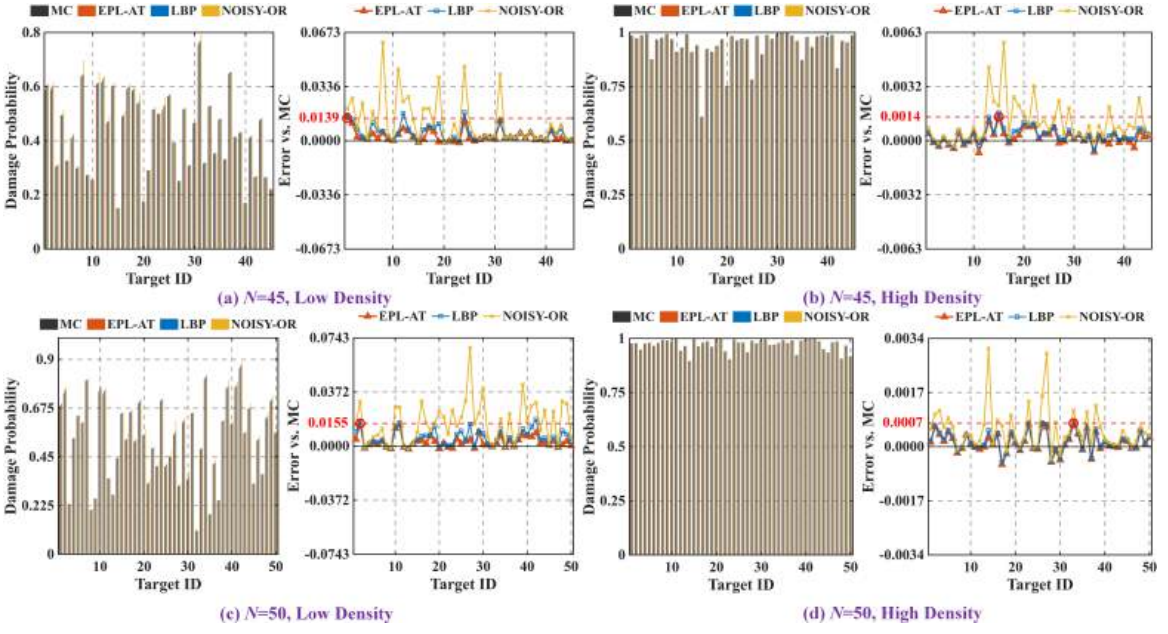


Figure 5: Comparison of damage probability estimations and absolute errors for large-scale scenarios ($N = 45$ and 50) under low and high network densities.

4.3. GS-MOPSO Comprehensive Verification

Convergence is a key indicator for evaluating algorithm performance and a decisive factor in measuring the quality of weapon-target assignment schemes. Since the true Pareto front of the DWTA problem is unknown, this paper evaluates the convergence of GS-MOPSO using the SPAD metric and conducts statistical analysis using the Wilcoxon rank-sum test and the Friedman test. All algorithms are run independently 30 times.

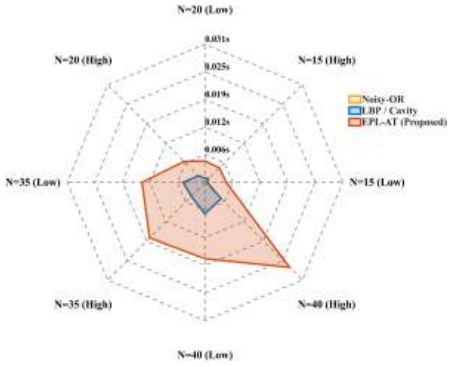


Figure 6: Radar chart evaluating the computational efficiency of different probability calculation models across varying target scales and densities.**(The average computation time of Monte Carlo is 0.2 s)**

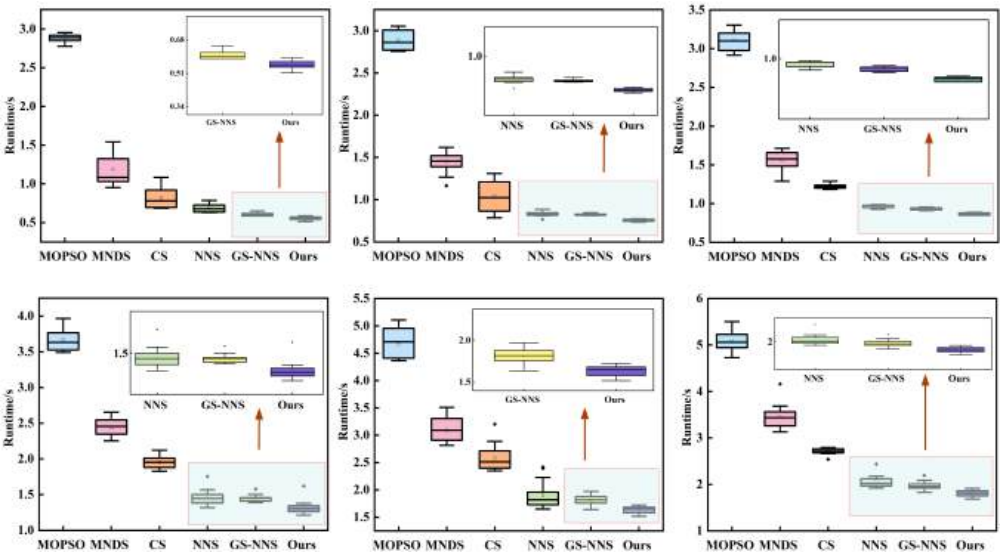


Figure 7: Runtime comparison of GS-MOPSO and other sorting strategies across six scenarios.

Specifically, the SPAD metric measures the convergence of an algorithm by calculating the Euclidean distance between the reference solution set R and the obtained Pareto solution set S (see Eq. (24)). The smaller the SPAD value, the better the algorithm's convergence. The compared algorithms include MOPSO, TC-MOPSO (Han et al. (2022)), ANMPSO (Gu et al. (2021)), APSEA (Tian et al. (2024)), and DVCEA (Ban et al. (2025)). Table 2 presents the SPAD comparison and statistical results for all algorithms. The results indicate that in most scenarios, GS-MOPSO exhibits a significant advantage in convergence. The statistical results further indicate that GS-MOPSO demonstrates the best convergence robustness.

$$SPAD(R,S) = \frac{\sum_{i=1}^{|R|} d_i^{\mathrm{SPAD}}}{|R|} \tag{24}$$

where $d_i^{\mathrm{SPAD}} = \min \|F(r^i) - F(x)\|, r^i \in R$; The reference solution set R selected in this paper consists of seven points, defined as $R = \{(0, \frac{1}{3}f_2^{\max}), (0, \frac{2}{3}f_2^{\max}), (0, f_2^{\max}), (0,0), (\frac{1}{3}f_1^{\max}, 0), (\frac{2}{3}f_1^{\max}, 0), (f_1^{\max}, 0)\}$.

To further evaluate the comprehensive performance of GS-MOPSO, this paper uses the HV metric (Guerreiro et al. (2021)) to assess its convergence and diversity. Table 3 presents the statistical results for all algorithms, which indicate that GS-MOPSO exhibits a significant advantage in computational performance. Furthermore, Fig. 8 presents the Pareto fronts corresponding to the median HV values obtained by all algorithms across six scenarios of different

Table 2
SPAD indicator results of the solution sets obtained by the algorithms

Scenario	MOPSO	TC-MOPSO	ANMPSO	APSEA	DVCEA	**GS-MOPSO**
Scenario 1	4.1785e-02 - (5.9301e-06)	4.0300e-02 = (2.3209e-06)	4.0054e-02 = (1.7985e-06)	4.0911e-02 = (1.9937e-06)	4.1224e-02 - (2.2307e-06)	**3.9900e-02** (1.1211e-06)
Scenario 2	2.5011e+02 - (1.3587e+04)	2.3104e+02 = (9.0812e+03)	2.4489e+02 - (8.2160e+03)	2.272251e-02 = (7.2154e+03)	2.2881e+02 = (5.3211e+03)	**2.2689e+02** (8.9444e+03)
Scenario 3	3.8591e-02 - (1.0684e-05)	3.8479e-02 - (9.4143e-06)	3.7846e-02 - (5.2502e-06)	3.7714e-02 - (2.0247e-06)	3.7739e-02 - (5.3266e-05)	**3.3616e-02** (4.6647e-06)
Scenario 4	3.8826e+02 - (2.7704e+04)	3.6790e+02 - (2.0645e+04)	3.4840e+02 - (2.7215e+04)	3.3577e+02 - (3.2473e+04)	3.2515e+02 = (2.6177e+04)	**3.2433e+02** (2.5659e+04)
Scenario 5	3.4517e-02 = (3.6233e-04)	3.5638e-02 - (5.2654e-04)	3.3329e-02 - (4.2245e-04)	3.2913e-02 = (3.1693e-04)	**3.0525e-02** + (3.2162e-04)	3.1872e-02 (4.3751e-04)
Scenario 6	3.4705e-02 - (3.2753e-04)	3.4419e-02 - (3.2542e-04)	**2.9828e-02** = (4.7518e-04)	3.3528e-02 - (6.5523e-04)	3.3236e-02 - (5.2575-04)	3.0235e-02 (3.7242e-04)
+ - =	0/5/1	0/4/2	0/4/2	0/3/3	1/2/3	

Table 3
HV indicator results of the solution sets obtained by the algorithms

Scenario	MOPSO	TC-MOPSO	ANMPSO	APSEA	DVCEA	GS-MOPSO
Scenario 1	4.9625e-01 = (5.3198e-06)	4.9784e-01 = (9.6694e-06)	4.9731e-01 = (3.9780e-06)	4.8962e-01 = (4.3215e-06)	4.7566e-01 - (5.7233e-06)	**4.9831e-01** (4.8599e-06)
Scenario 2	3.7704e-01 - (3.9984e-05)	3.8294e-01 - (2.9802e-05)	3.8336e-01 - (3.1112e-05)	3.8257e-01 = (3.2543e-05)	3.8859e-01 = (3.6565e-01)	**3.9384e-01** (2.2551e-05)
Scenario 3	5.7898e-01 - (9.9544e-05)	5.9222e-01 = (9.2149e-05)	5.8942e-01 - (9.8665e-05)	5.7665e-01 - (5.6571e-05)	5.8551e-01 - (7.3312e-05)	**5.9256e-01** (8.1679e-05)
Scenario 4	4.6511e-01 - (1.2138e-04)	4.6836e-01 - (1.1272e-04)	4.6932e-01 - (1.7373e-04)	4.6851e-01 - (1.6588e-04)	**4.8209e-01** = (2.0456e-04)	4.7219e-01 (1.1714e-04)
Scenario 5	4.5677e-01 - (5.7531e-04)	4.5262e-01 - (6.0153e-05)	4.7119e-01 - (3.2713e-05)	4.8027e-01 - (5.0237e-05)	4.7579e-01 - (6.3222e-05)	**4.9011e-01** (5.2532e-05)
Scenario 6	5.0256e-01 - (4.3126e-04)	5.1338e-01 - (5.7244e-05)	5.2105e-01 - (6.0702e-05)	5.3011e-01 - (4.3271e-05)	5.2999e-01 - (4.5237e-05)	**5.4007e-01** (3.7588e-05)
+ - =	0/5/1	0/4/2	0/5/1	0/4/2	0/4/2	

scales. The illustrated results further confirm this point. In addition, the Friedman test results shown in Fig. 9 indicate that GS-MOPSO ranks first in terms of computational performance.

In terms of runtime, the efficiency ranking radar chart of GS-MOPSO and the five compared algorithms is shown in Fig. 10. Clearly, GS-MOPSO, which integrates all proposed strategies, exhibits the best computational efficiency across all scenarios.

5. Conclusion

This article addresses the issue of overestimation in the calculation of target damage probability using traditional DWTA models. Cavity probability theory is introduced to propose the CF-DWTA model, and an EPL-AT truncation strategy is designed to effectively eliminate redundant probabilities in complex correlation networks. To solve the model, GS-MOPSO is designed to improve the computational efficiency and convergence performance of the algorithm by optimizing the population initialization strategy, particle sorting strategy, and archive maintenance mechanism. The simulation results under different scale scenarios show that the CF-DWTA model can more accurately calculate the probability of target damage; meanwhile, GS-MOPSO outperforms the compared algorithms in terms of SPAD, HV, and Runtime metrics.

To further approximate the complex battlefield environment in reality, we will study the assignment of heterogeneous defense resources under highly dynamic topology and uncertain conditions in the future. In addition, introducing data-driven methods to meet the real-time solving needs in large-scale swarm attack scenarios is also an important direction for our future research.

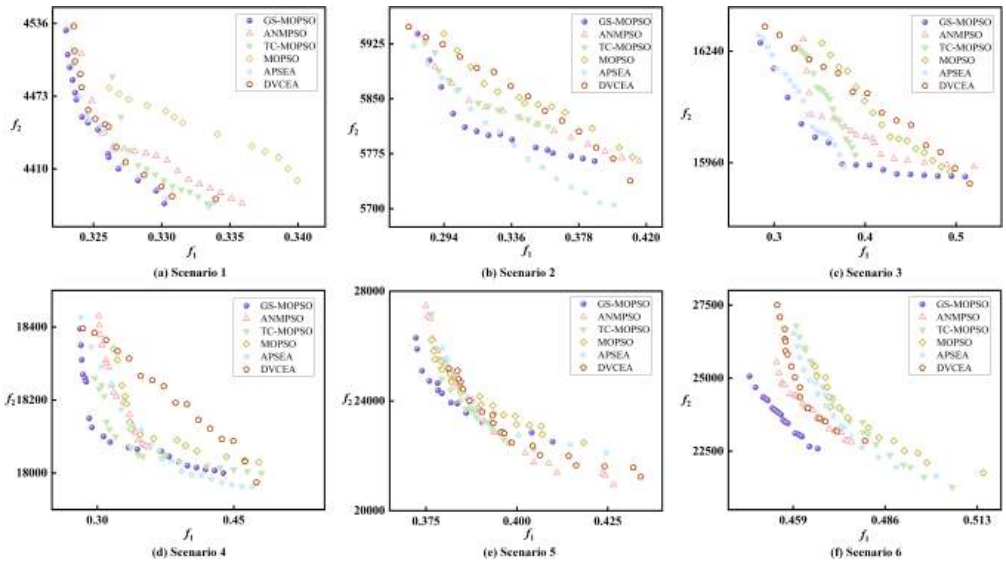


Figure 8: Pareto fronts corresponding to the median HV values obtained by the six compared algorithms across six different scale scenarios.

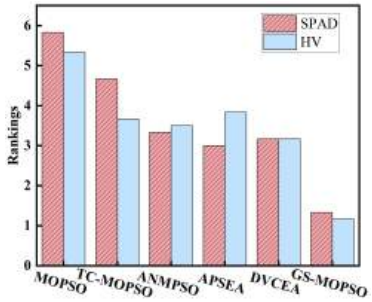


Figure 9: Average rankings of the compared algorithms based on SPAD and HV indicators using the Friedman test.

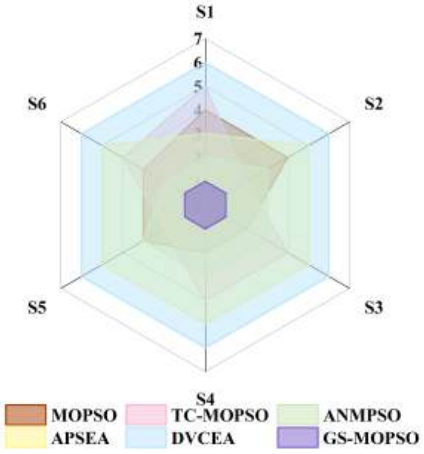


Figure 10: Radar chart illustrating the comprehensive performance rankings of all algorithms across the six simulation scenarios.

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